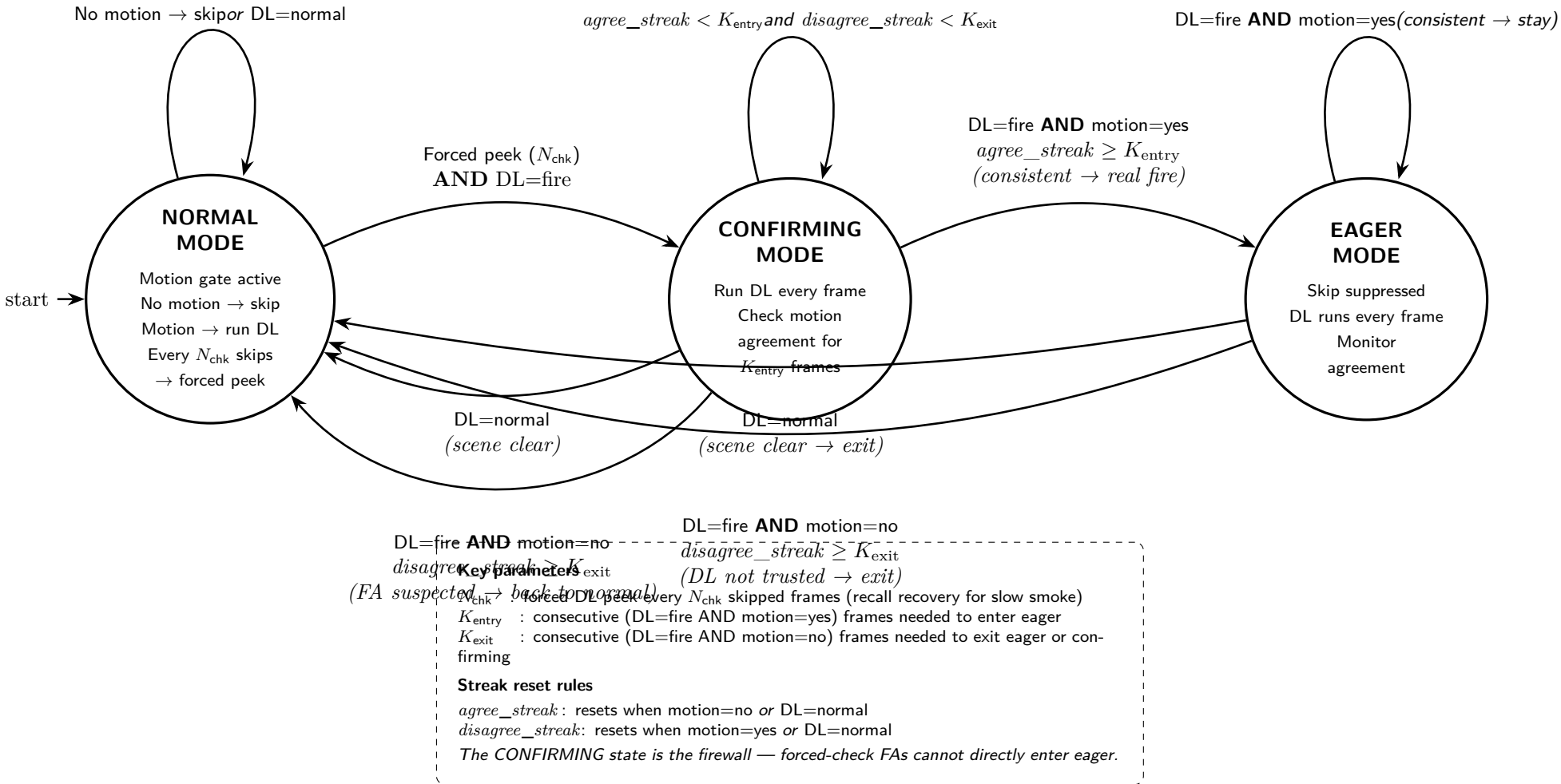
No motion → skipor DL=normal
agree_streak < K_entry and disagree_streak < K_exit
DL=fire AND motion=yes(consistent → stay)
Forced peek (N_chk)
AND DL=fire
DL=fire AND motion=yes
agree_streak ≥ K_entry
(consistent → real fire)
NORMAL MODE
Motion gate active
No motion → skip
Motion → run DL
Every N_chk skips
→ forced peek
CONFIRMING MODE
Run DL every frame
Check motion
agreement for
K_entry frames
EAGER MODE
Skip suppressed
DL runs every frame
Monitor
agreement
start
DL=normal
(scene clear)
DL=normal
(scene clear → exit)
DL=fire AND motion=no
disagree_streak ≥ K_exit
(DL not trusted → exit)
DL=fire AND motion=no
disagree_streak ≥ K_exit
(FA suspected → back to normal)
Key parameters
N_chk : forced DL peek every N_chk skipped frames (recall recovery for slow smoke)
K_entry : consecutive (DL=fire AND motion=yes) frames needed to enter eager
K_exit : consecutive (DL=fire AND motion=no) frames needed to exit eager or confirming
Streak reset rules
agree_streak : resets when motion=no or DL=normal
disagree_streak: resets when motion=yes or DL=normal
The CONFIRMING state is the firewall — forced-check FAs cannot directly enter eager.